FINM3013 Introduction to Financial Derivatives

Chapter 11

Solution to Class exercises

1. A. A bull spread is created by buying a low strike call and selling a high strike call. Alternatively, it can be created by buying a low strike put and selling a high strike put.

2. C. The gain from a very high stock price or a very low stock price is the same. Suppose calls are used. In the case of a very low stock price none are exercised and the gain is $c_1+c_3-2c_2$ from the option premium. In the case of a very high stock price all options are exercised. The net payoff is zero and the gain is the same.

3. D.

4. A. A protective put consists of a long put plus the stock. The holder of the put owns the stock that might become deliverable.

5. B. A one-year zero-coupon bond is worth $1000e^{-0.05\times 1}$ or about \$951. This leaves $1000-951 = \$49$ for buying the option.